Math Daily Log

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acknowledgment

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October 28, 2025

1. Weyl Algebra $\frac{k<x,y>}{xy-yx-1}$.

2. Center of Weyl Algebra is k in characteristic 0.

3. In characteristic p, the center of Weyl Algebra is $k[x^p, y^p]$.

4. Weyl Algebra is simple.

5. x and y can be seen as operators on $k[t]$ by $x : f \mapsto tf$ and $y : f \mapsto \frac{df}{dt}$. This is amazing. I think the structure $xy - yx = 1$ can be used to make some algebra olympiad problems. Very rich and interesting structure. When you interpret as operators on polynomials, everything becomes very concrete and easy to understand. Perfect for a math olympiad problem.

October 29, 2025

1. Splitting field of a polynomial is unique up to isomorphism.

2. Separable. An inseparable polynomial: $x^p - t$ over $\mathbb{F}_p(t)$. $x^p - t = (x - u)^p$ in the algebraic closure.

3. A polynomial is separable iff its derivative and f are coprime. In particular, over characteristic 0, all irreducible polynomials are separable.

4. A field is perfect if all irreducible polynomials are separable iff every algebraic extension is separable.

5. A finite extension is separable iff $[F : k]_s = [F : k]$ iff $F = k(\alpha_1, ..., \alpha_n)$ where α_i are separable.

6. A finite Normal extension is a splitting field of a polynomial. This is amazing.

7. Riemann integral is equivalent to Darboux integral.

October 30, 2025

1. Riemann integrable iff discontinuities has measure 0.

2. A lot of exercises on separable extensions from Aluffi chapter 7.

3. Formal power series $k[[x]]$ on math 250 lecture.

Nov. 1, 2025

1. Cyclotomic polynomial is in $\mathbb{Z}[x]$,and it's always irreducible. Proof used Dirichlet's Theorem and the fact that finite fields are perfect. Too smart.

2. Finite Field of order p^r is unique up to isomorphism, it's the splitting field of $x^{p^r} - x$.

3. $x^4 + 1$ is reducible in all finite field. But irreducible in $\mathbb{Z}[x]$. Proof use the fact that $x^4 + 1 \mid x^8 - 1 \mid x^{p^2} - x$. So the root β of $x^4 + 1$ in F_{p^2} has degree of 1 or 2. Thus, its minimal polynomial has degree of 1 or 2.

4. Let $F = \mathbb{F}_q$, $x^{p^n} - x$ is the product of all the irreducible polynomial of degree d for all $d \mid n$.

5. The automorphism $Aut_{\mathbb{F}_p}(\mathbb{F}_{p^d}) \cong C_q$. The generator is Frobenius function $x \mapsto x^p$.

Nov. 2, 2025

1. There are countably many jump discontinuities.

2. Set of Critical Value is zero set.

3. The automorphism $\phi \in Aut_k(k(x))$ must be $\phi : x \mapsto \frac{ax+b}{cx+d}$ where $ac-bd \neq 0$. A surprising result but hard to prove. Is there a natural but easy way to see this?

4. Review H104.

Nov. 3, 2025

1. $k \subseteq F$ iff there are finitely many fields E such that $k \subseteq E \subseteq F$. This is highly nontrivial to me. My intuition is that if $k(a, b)$ is not a simple extension of k, then it may have a lot of possible combination of a and b to form an intermediate field E. This turns out to be helpful for one direction.

2. All finite separable extensions are simple. This is very surprising and makes our life substantially easier. Consequently, it's very rare to see a nonsimple extension. The proof is very hard, and I find it hard to come up with.

Nov. 4, 2025

1. Set up Wolfram.

2. Weddernburn Little Theorem: all finite division rings are fields.

3. A Riemann integrable function has integral 0 iff it's zero almost everywhere (set of roots is a zero set).

4. Antiderivative Theorem: The antiderivative of a Riemann integrable function, if it exists, differs from the indefinite integral by a constant. This can be proved using Meaan Value Theorem.